Lecture 1: Non-cooperative game theory

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Lecture 1 Outline

- Game theory basics
- Nash equilibrium
- Sub-game perfect Nash equilibrium
- Extra Reading: any entry level textbooks like
  - Gibbons (1992) Game theory for applied economists. Princeton University Press.
  - Dixit, Skeath and Reiley (2015) Games of strategy. WW Norton & Company.

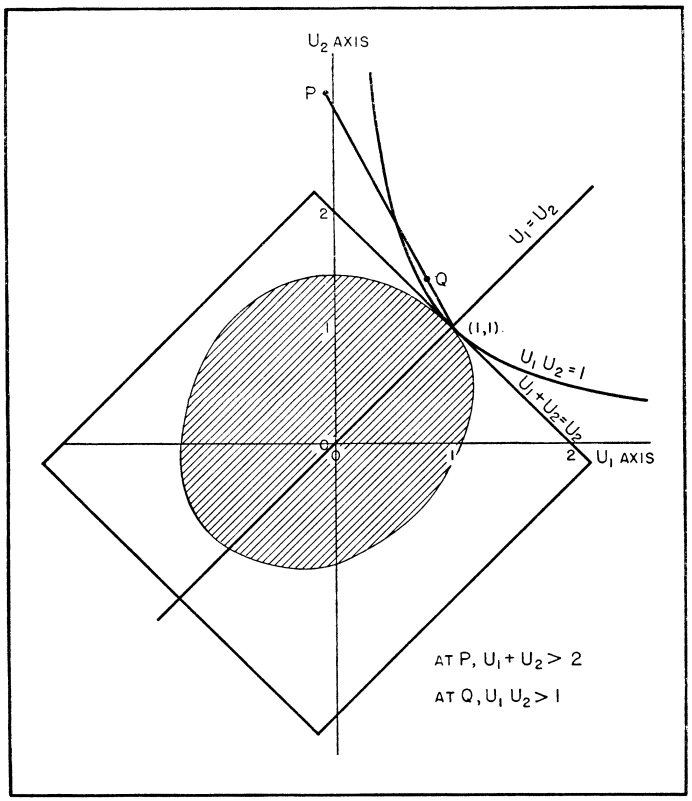
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What is Game Theory?

- **Interactions**
- A study of strategically interdependent behavior
- What I do affects you and what you do affects me
- A natural way of thinking, something you already know and master (Video)

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How to study Game Theory?



I assume that the preferences satisfy the following five assertions:

For all $r, s \in S, t, t_1, t_2 \in N$, and $i \in \{1, 2\}$:

(A-1) if $r_i > s_i$, then $(r, t) >_i (s, t)$;

(A-2) if $s_i > 0$ and $t_2 > t_1$, then $(s, t_1) >_i (s, t_2) >_i (0, \infty)$;

(A-3) $(r, t_1) \gtrsim_i (s, t_1 + 1)$ iff $(r, t_2) \gtrsim_i (s, t_2 + 1)$;

(A-4) if $r_n \to r$ and $(r_n, t_1) \gtrsim_i (s, t_2)$, then $(r, t_1) \gtrsim_i (s, t_2)$;
if $r_n \to r$ and $(r_n, t_1) \gtrsim_i (0, \infty)$, then $(r, t_1) \gtrsim_i (0, \infty)$;

(A-5) if $(s + \epsilon, 1) \sim_i (s, 0), (\bar{s} + \bar{\epsilon}, 1) \sim_i (\bar{s}, 0)$, and $s_i < \bar{s}_i$, then $\epsilon_i \leqq \bar{\epsilon}_i$.

From (A-3) we can use the notation $(r, T) \gtrsim_i (s, 0)$ and $(r, T) \lesssim_i (s, 0)$ for $(r, T + t) \gtrsim_i (s, t)$ and $(r, T + t) \lesssim_i (s, t)$, respectively.

Two families of models in which the preferences satisfy the above conditions are:

I. *Fixed bargaining costs*. Each player i has a number c_i such that $(s, t_1) \gtrsim (\bar{s}, t_2)$ iff $(s_i - c_i \cdot t_i) \geqq (\bar{s}_i - c_i \cdot t_2)$.

II. *Fixed discounting factors*. Each player i has a number $0 < \delta_i \leqq 1$ such that $(s, t_1) \gtrsim_i (\bar{s}, t_2)$ iff $s_i \delta_i^{t_1} \geqq \bar{s}_i \delta_i^{t_2}$.

We reserve $\delta_i = 0$ for the lexicographic preference: $(s, t_1) \gtrsim_i (\bar{s}, t_2)$ if $(t_1 < t_2)$ or $(t_1 = t_2$ and $s_i \geqq \bar{s}_i)$.

REMARK: In a more general framework of the model, player i would be characterized by the sequence of preferences $\{\gtrsim_i^t\}$, where $\gtrsim_i^t$ is i's preference on the outcomes assuming that the players have not reached an agreement in the first $t - 1$ periods. In fact, I assume that $\gtrsim_i^t \equiv \gtrsim_i$. This assumption precludes discussion of some interesting bargaining situation such as: (1) player i has a sequence $\{c_i^t\}$ where c_i^t is the cost to i of the bargaining in period t; (2) player i has a fixed bargaining cost and his utility is not linear.

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How to study Game Theory?



How to study Game Theory?

- **Interactive thinking** & logical reasoning
- Put yourself in other's shoes
- Think what others think
- Try to teach game theory

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Basic assumption: Rationality

- Rationality
  - Selfish: caring only about one's own benefit. No fairness, no revenge, no nothing.
  - Smart: able to solve the unsolvable.
- Example: Guessing game
  - Now, everyone guesses a number from 0 to 100.
  - Take the average of everyone's guess.
  - Take the half of the average.
  - The person whose guess is the closest to the half of the average wins 10000000000 RMB.
  - Level k thinking

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Basic assumption: Common Knowledge

- Common Knowledge
  - I know ...
  - I know that you know ...
  - I know that you know that I know ...
  - I know that you know that I know that you know ...
  - .....
  - Chuang-tzu and fish (Video)
- Example: A simple secret, say, I'm broke
  - Scenario 1: tell Peppa and Peter in private, respectively
  - Scenario 2: tell Peppa and Peter in public, loudly

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Elements of a game

Players	Who are playing?
Actions	What are their choices?
Payoffs	What outcomes would the choices lead to?



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Different games

- Non-cooperative/Cooperative games
- Static/Dynamic games
- Perfect/Imperfect information games
- Complete/Incomplete information games
- Zero-sum/non zero-sum games
- Behavior games
- Evolutionary games
- ...

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Chatgpt to help...(optional)

what is thedifference between incomplete information and imperfect information?

Incomplete information and imperfect information are related concepts, but there is a subtle difference between them.

Incomplete information refers to a situation where some relevant information is missing or unknown to one or more parties in a decision-making process. This can occur in situations where information is too costly to obtain or where some information is kept secret or intentionally withheld. Incomplete information can lead to uncertainty and ambiguity in decision-making and can make it difficult to make optimal decisions.

Imperfect information, on the other hand, refers to a situation where the available information is not fully reliable or accurate. This can occur due to factors such as asymmetric information (where one party has more information than the other), hidden actions or hidden characteristics, or moral hazard (where one party may take actions that are not in the best interest of the other party). Imperfect information can lead to suboptimal decisions, as the decision-maker may be working with inaccurate or incomplete information.

In summary, incomplete information refers to a lack of relevant information, while imperfect information refers to information that is unreliable or inaccurate. Both incomplete and imperfect information can create challenges for decision-making, and various tools and techniques (such as Bayesian analysis and game theory) are used to help address these challenges.



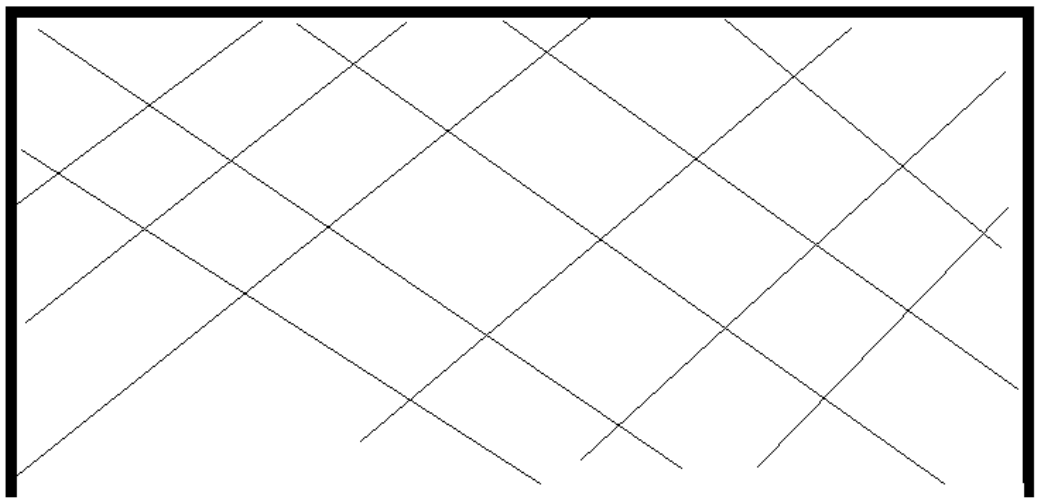
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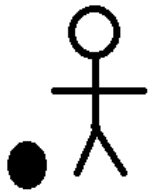
Applications of Game Theory



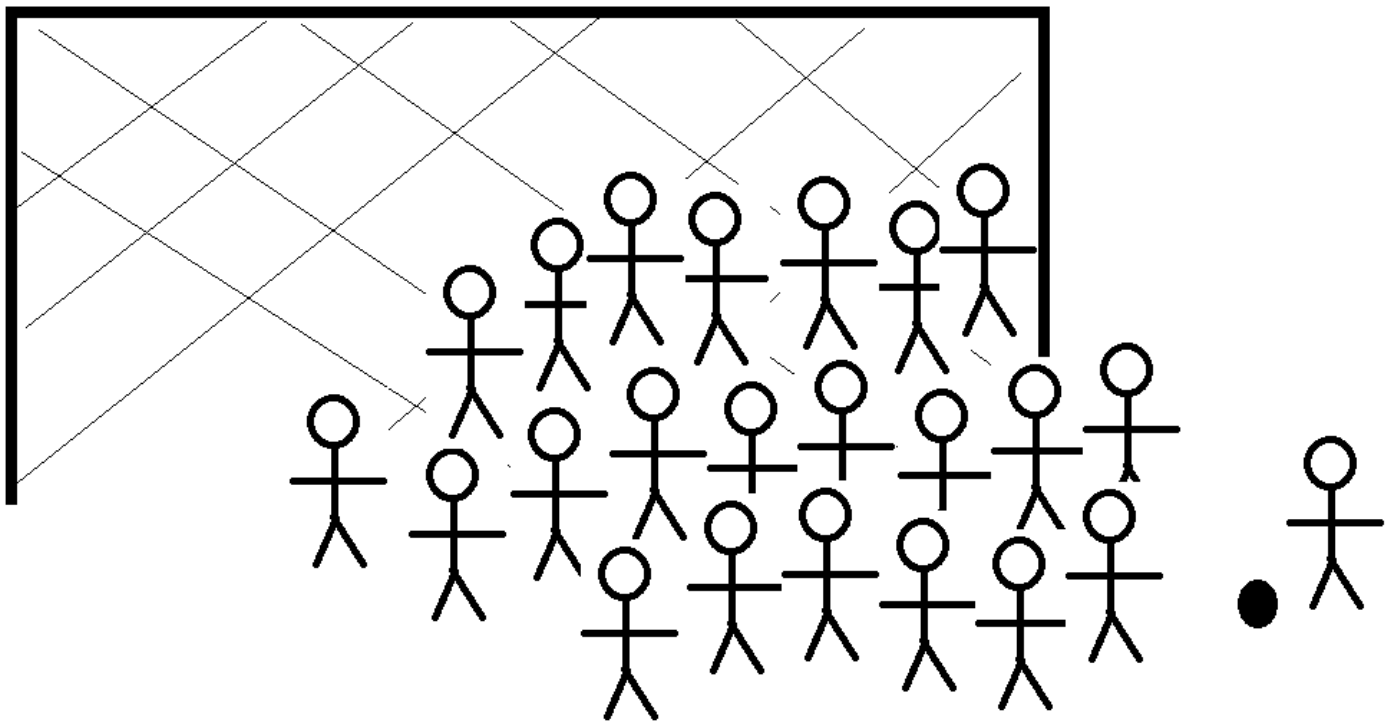
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Application in sport: open goal

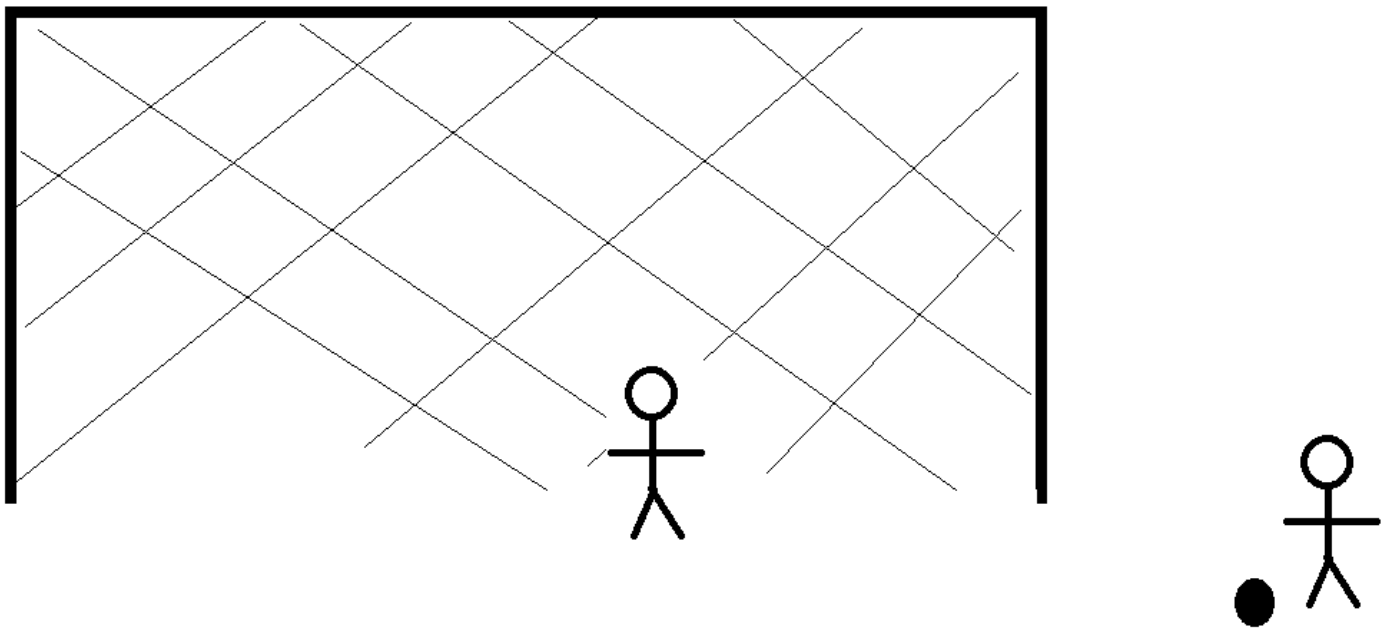




Application in sport: free kick



Application in sport: penalty goal



Application in Sport: summary

1 player	Open Goal	Individual Optimization, Monopoly
n players $n = 2, 3, 4...$	Penalty Kick	Bargaining, Auction, Duopoly, Oligopoly...
∞ players	Free Kick	Perfect competition

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Nash Equilibrium

Definition

- A **state** in which **no player** has an **incentive** to **unilaterally** change his **current strategy**. (Video)

About NE...

- Equilibrium and Nash equilibrium
- Whether it is an appropriate prediction
- The "stability"
- Why so important?
- Examples: Solow model, perfect competition, Rock-paper-scissors, Seat taking, etc...

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Nash Equilibrium

Derivation

- Dominant strategy
- Iterative elimination of dominated strategies
- Best responses analysis (pure strategy NE)
- Opponent's indifference property (mixed strategy NE)

Pure and mixed strategy NE

- Shouldn't pure strategy be a special type of mixed strategy, with $p = 0$ and $q = 1$ for example? Why not?

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Nash Equilibrium

Two "no incentive to deviate":

- (In a certain status, given other players' current strategies) each player will be worse if he deviates.
- (In a certain status, given other players' current strategies) each player will be indifferent if he deviates.



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Example

		Player 2	
		Left	Right
Player 1	Up	(7,8)	(1,2)
	Down	(3,4)	(5,6)

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Example

		Player 2	
		Left	Right
Player 1	Up	(7,8)	(1,2)
	Down	(3,4)	(5,6)

		Player 2	
		Left ($q = \frac{1}{2}$)	Right ($1 - q = \frac{1}{2}$)
Player 1	Up ($p = \frac{1}{4}$)	($\underline{7}$,$\underline{8}$)	(1,2)
	Down ($1 - p = \frac{3}{4}$)	(3,4)	($\underline{5}$,$\underline{6}$)

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Example

		Player 2	
		Left	Right
Player 1	Up	(7,8)	(1,2)
	Down	(3,4)	(5,6)

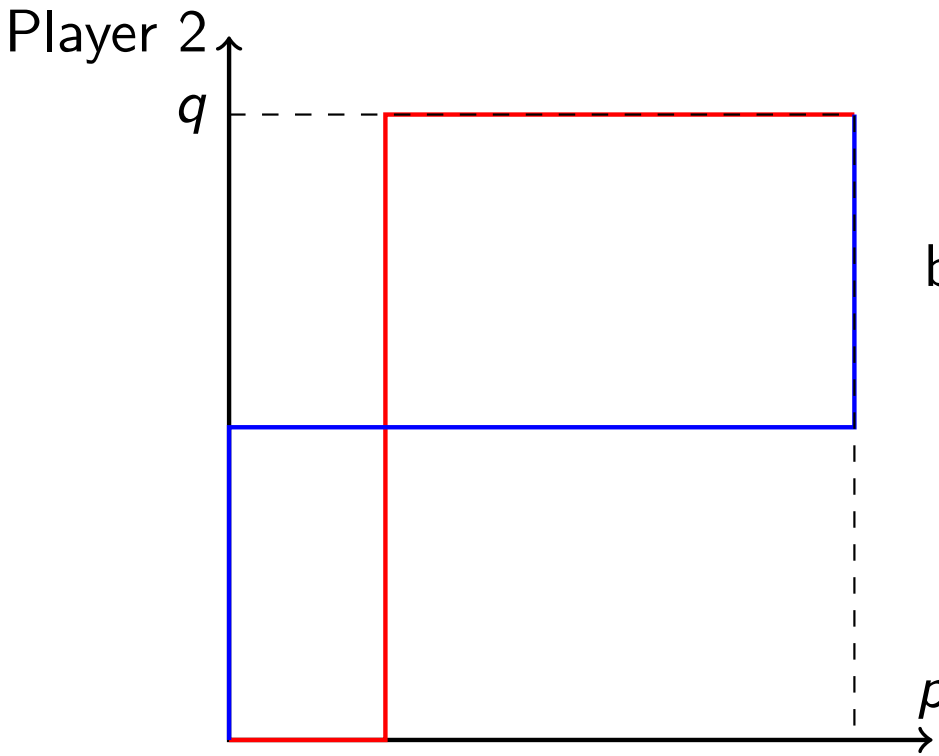
- 1 chooses Up: $7q + (1 - q)$
- 1 chooses Down: $3q + 5(1 - q)$
- 1 chooses UP with p and Down with $1 - p$: $p(7q + (1 - q)) + (1 - p)(3q + 5(1 - q))$
- 2 chooses Left: $8p + 4(1 - p)$
- 2 chooses Right: $2p + 6(1 - p)$
- 2 chooses Left with q and Right with $1 - q$: $q(8p + 4(1 - p)) + (1 - q)(2p + 6(1 - p))$



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Example

- 1's best response function: $p = 1$ if $q > \frac{1}{2}$, $p = 0$ if $q < \frac{1}{2}$, indifference if $q = \frac{1}{2}$.
- 2's best response function: $q = 1$ if $p > \frac{1}{4}$, $p = 0$ if $q < \frac{1}{4}$, indifference if $p = \frac{1}{4}$.



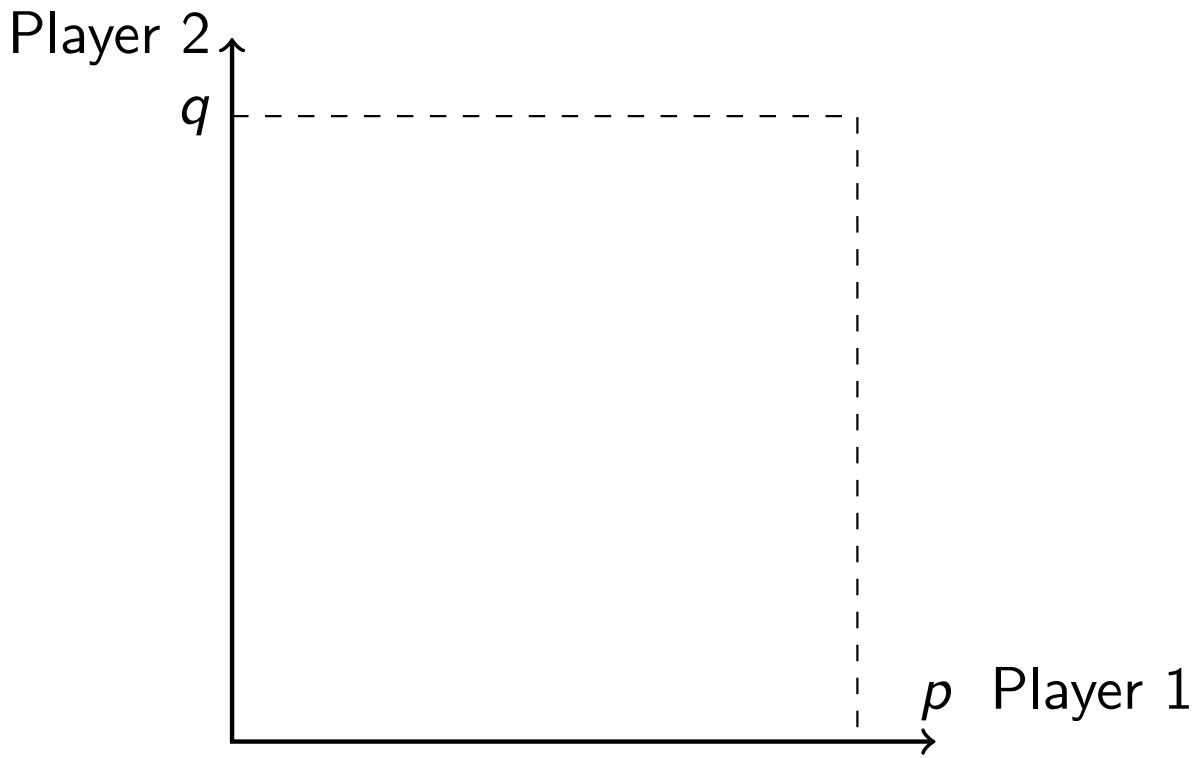
blue: 1's best response function

red: 2's best response function

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Existence of NE (optional)

- Fixed point theorem (Gibbons Chapter 1)
- Try to draw a "continuous" best response curve from player 2 to player 1, and a "continuous" best response curve from player 1 to player 2. Don't let them intersect.



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Sub-game perfect Nash Equilibrium (SPE)

Definition

- A **state** in which **no player** has an **incentive** to **unilaterally** change his **current strategy** in **every node** of the game.

Derivation

- Forward looking
- Backward induction (if the horizon is finite)

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Basic assumption in bargaining

- For convenience, assume player 1 as male and player 2 as female.
- In each negotiation, assume the total pie to be 1.
- When player 1 is proposing, he offers a share of x to player 2, leaving $1 - x$ to himself. Similarly, when player 2 is proposing, she offers a share of y to player 1, leaving $1 - y$ to herself.



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Ultimatum game: a simple practice on SPE

- You're already familiar with the ultimatum game.
- Player 1 makes an offer x to player 2, player 2 could accept x and that is the end of the game. If player 2 rejects x then both player earns a zero payoff. What is the SPE?
- In the second stage: 2 chooses between accepting and rejecting every x she might receive. Clearly, she would surely accept any $x > 0$.
- Anticipating 2's response, 1 would make the smallest positive offer as possible. In the extreme case, 1 offers $x = 0$.
- The full description of SPE: 1 offers $x = 0$, 2 accepts any $x \geq 0$.



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Ultimatum game: let's torture our minds (optional)

- Further think: Shouldn't player 2 be indifferent between accepting and rejecting $x = 0$?
- Suppose that player 2 accepts $x = 0$ with a probability less than 1, or, she MIGHT reject $x = 0$. Anticipating that, player 1 would slightly increase the offer x by ϵ. However, no matter how small ϵ is, he has an incentive to further reduce it. This process goes on and on and no SPE exists.
- Therefore, the only SPE that could survive is the one in which player 1 offers $x = 0$ and player 2 accepts for sure.
- Seems tricky: looks like we find this SPE not because it makes the most sense, but because SPE is so poorly defined that it takes away other choices that make sense.
- To make our life easier, we have the following Tie-breaking rule: whenever a player is indifferent between choices, he (she) chooses the one that benefits the other player.



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Finite horizon Alternating-offer Bargaining model (optional)

- Starting from the ultimatum game, we can easily imagine a T stages, finite horizon alternating-offer bargaining model. In each stage, 1 offers x to 2 or 2 offers y to 1. Without loss of generality, assume that player 1 is the final proposer, and there is a discount factor δ.
- In stage T, player 2 accepts iff $x_T \geq 0$, player 1 offers $x_T = 0$ and earns $1 - x_T = 1$.
- In stage $T-1$, player 1 accepts iff $y_{T-1} \geq \delta$, player 2 offers $y_{T-1} = \delta$ and earns $1 - y_{T-1} = 1 - \delta$.
- In stage $T-2$, player 2 accepts iff $x_{T-2} \geq \delta(1-\delta)$, player 1 offers $x_{T-2} = \delta(1-\delta)$ and earns $1 - x_{T-2} = 1 - \delta(1-\delta)$
- In stage $T-3$, player 1 accepts iff $y_{T-3} \geq \delta(1-\delta(1-\delta))$, 2 offers $y_{T-3} = \delta(1-\delta(1-\delta))$ and ...



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Finite horizon Alternating-offer Bargaining model (optional)

- Therefore, the minimum acceptable offers in every period of the game are
  - $x_T = 0$
  - $y_{T-1} = \delta$
  - $x_{T-2} = \delta - \delta^2$
  - $y_{T-3} = \delta - \delta^2 + \delta^3$
  - $x_{T-4} = \delta - \delta^2 + \delta^3 - \delta^4$
  - ...
  - $x_t = y_t = \delta \frac{1-(-\delta)^{T-t}}{1+\delta}$
- When T approaches infinity, the minimum acceptable offer is $\frac{\delta}{1+\delta}$.

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Thank you



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